

5. (a) Define the *work function* of a metal.

[1]

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- (b) The work function of caesium is 3.2×10^{-19} J. Show that the frequency of light that will eject electrons from a caesium surface with a maximum kinetic energy of 1.5×10^{-19} J is approximately 7×10^{14} Hz.

[2]

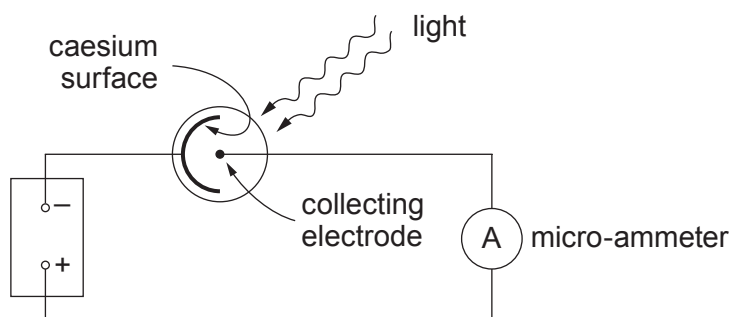
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- (c) The **same frequency** of light is shone on to the caesium surface in a vacuum photocell in the circuit shown.



The light energy falling each second on the caesium surface is 0.30 J.

- (i) Show that the number of photons striking the caesium surface each second is approximately 6×10^{17} .

[2]

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- (ii) The current indicated by the ammeter is $0.80\,\mu\text{A}$. Calculate the number of electrons per second emitted from the caesium surface, stating your assumption. [3]

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- (iii) Hence calculate the *probability* of a photon of this frequency ejecting an electron from a caesium surface. [1]

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